The Thermodynamic Floor of Vertebrate Cellular Identity: Methylation Entropy, Body Temperature, and Maximum Lifespan across 500 Million Years of Evolution

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Abstract

We report that the methylation entropy floor of somatic cellular identity, derived from first thermodynamic principles in the Informational Actualization Model (IAM) [Mahaffey, 2026], predicts maximum lifespan across all jawed vertebrates with a correlation of $r = -0.90$ ($p = 1.6 \times 10^{-16}$, $n = 43$ species). Long-lived K-selected species — bowhead whale (211 yr), human (122 yr), African elephant (70 yr) — maintain methylation entropy near the thermodynamic floor ($A \approx 0.98$–1.00). Short-lived r-selected species — house mouse (4 yr), Norway rat (5 yr), common shrew (2.5 yr) — operate significantly above it ($A \approx 1.12$–1.16). The 105-fold lifespan difference between shrew and bowhead whale corresponds to a $\Delta A = 0.178$ in methylation entropy, with zero free parameters. Extending the framework to all jawed vertebrates (fish, amphibians, reptiles, birds) via a body-temperature correction $A(T) = H(\beta)/[H_{\min} \times (T_{\text{body}}/310.15\,\text{K})^{\alpha}]$ with empirically derived $\alpha = 2.0$ reduces cross-class variance by 41%. The framework boundary is precisely defined: it applies wherever DNMT1-mediated CpG maintenance methylation is the primary epigenomic identity mechanism — all jawed vertebrates, approximately 50,000 species — and does not apply where this mechanism is absent or vestigial (arthropods, nematodes). These results suggest that lifespan is thermodynamically constrained by the fidelity cost of maintaining cellular identity, and that the Landauer entropy floor is a universal law of jawed vertebrate biology, conserved across 500 million years of evolution.

Keywords: DNA methylation; Shannon entropy; Landauer principle; thermodynamics; lifespan; vertebrate evolution; cellular identity; aging; epigenetic clock; body temperature

1 Introduction

Why do mammals live so differently? The house mouse and the bowhead whale share approximately 80% of their protein-coding genes [Waterston et al., 2002]. Their cellular machinery is nearly identical at the molecular level. Yet one lives four years; the other, two centuries.

Epigenetic clocks have documented a deep connection between DNA methylation and lifespan. Lowe et al. [2018] showed that the *rate* of methylation change at conserved age-related

CpG sites scales with maximum lifespan across 42 mammalian species. Lu et al. [2023] demonstrated that universal pan-mammalian clocks trained on 11,754 arrays from 185 species achieve $r > 0.96$ accuracy for chronological age estimation. Haghani et al. [2023] characterized the co-methylation networks underlying mammalian traits across 167 eutherian species. The empirical pattern is robust: methylation tracks lifespan.

What has been missing is a *physical explanation* for why this is so. These clocks are statistical: trained regression models that identify CpGs correlated with age. They accurately describe what happens. They do not derive from first principles why it happens.

The Informational Actualization Model (IAM) provides that derivation. IAM applies the Landauer principle [Landauer, 1961] to the DNMT1-mediated CpG maintenance reaction: each correct maintenance event during DNA replication writes one bit of epigenomic information, dissipating a minimum energy $k_B T_{\text{body}} \ln 2$ as heat. From this, IAM derives a minimum Shannon entropy H_{min} below which no cell of a given architectural class can sustain its epigenomic identity through cell division. The dimensionless epigenomic fidelity index $A = H(\beta)/H_{\text{min}}$ measures how far a cell's actual methylation entropy lies above this thermodynamic floor.

In the companion paper [Mahaffey, 2026], we validated H_{min} for eight mammalian cell architecture classes using MCMC against 37 published reference cell measurements (17 chains, $\hat{R} < 1.001$), and demonstrated zero-free-parameter prediction of cancer floor departure in 27/28 TCGA cancer types. Here we extend that framework to ask: does the same thermodynamic floor explain why different species live different lengths of time?

The answer, we show, is yes.

2 Methods

2.1 The A-score and thermodynamic floor

The Shannon binary entropy of a methylation beta value β is:

$$H(\beta) = -\beta \log_2 \beta - (1 - \beta) \log_2 (1 - \beta) \tag{1}$$

The epigenomic fidelity index is:

$$A = \frac{H(\beta)}{H_{\text{min}}(\text{class})} \tag{2}$$

where H_{min} is the G-002 MCMC posterior for the relevant architecture class ($H_{\text{min}}^{\text{immune}} = 0.838889 \pm 0.000681$, 17 chains, all blood measurements use this value).

For blood samples, the immune cell class ($H_{\text{min}} = 0.838889$) is the appropriate reference, as immune cells constitute approximately 70% of circulating cell-free DNA [Moss et al., 2018].

2.2 Mammalian dataset (VAL-034)

We compiled published mean global methylation beta values for 40 mammalian species from three sources: Lowe et al. [2018] (42 species, blood, conserved mammalian CpG panel), Lu et al. [2023] (supplementary data, 185 species), and Wang et al. [2020] (104 Labrador retrievers, age-stratified cohort). All beta values are from healthy adult blood drawn post-sexual maturity, consistent with the protocol in Lu et al. [2023].

For the lifespan analysis, we used published maximum lifespans from AnAge database [Tacutu et al., 2018]. Body mass was obtained from PanTHERIA [Jones et al., 2009].

2.3 Vertebrate extension (VAL-035)

We extended the analysis to non-mammalian vertebrates using published global CpG methylation measurements from Varriale and Bernardi [2006a,b] (fish, reptiles, mammals, birds comparison via HPLC), Feng et al. [2010] (zebrafish, 79.5% global methylation), Bertucci et al.

[2021] (brown anole lizard epigenetic clock), and Lyko [2018] (honey bee insect methylation). Body temperatures were obtained from primary species biology literature.

For ectotherms (reptiles, amphibians, fish), we propose a body-temperature correction to the thermodynamic floor:

$$H_{\min}(T) = H_{\min}(37°\mathrm{C}) \times \left(\frac{T_{\text{body}}}{310.15\,\mathrm{K}}\right)^{\alpha} \tag{3}$$

The exponent α is empirically derived by minimizing the cross-class variance of temperature-corrected A values across all jawed vertebrates. The physical motivation is that the Landauer cost scales with T_{body}: lower temperature reduces the per-bit maintenance cost, allowing cells to sustain higher-entropy (higher-β) states at equilibrium.

2.4 Statistical analysis

Pearson and Spearman correlations between log(maximum lifespan) and A were computed using `scipy.stats`. Significance thresholds: $p < 0.05$ (nominal), $p < 0.001$ (strong). All scripts are reproducible and archived at the Zenodo DOI [Mahaffey, 2026].

3 Results

3.1 Lifespan predicts methylation entropy across mammals

Figure 1 shows the relationship between maximum lifespan and A across 40 mammalian species spanning 14 taxonomic orders. The correlation is strong and highly significant: Pearson $r = -0.9018$ ($p = 1.6 \times 10^{-16}$), Spearman $\rho = -0.9149$ ($p = 9.5 \times 10^{-18}$).

The pattern is taxonomically coherent:

Table 1: A by taxonomic order, sorted by mean lifespan (mammalian dataset)

Order	N	Mean A	σ	Mean lifespan (yr)	Interpretation
Cetacea	5	0.997	0.016	99	At thermodynamic floor
Proboscidea	1	0.987	–	70	At floor
Primates	6	1.007	0.015	49	Slightly above floor
Artiodactyla	4	1.015	0.011	30	Slightly above floor
Chiroptera	4	1.041	0.007	35	Above floor (lifespan outliers for size)
Carnivora	9	1.053	0.023	28	Above floor
Lagomorpha	2	1.114	0.002	8	Well above floor
Rodentia	6	1.125	0.018	9	Well above floor
Insectivora	1	1.157	–	2	Furthest from floor

The bimodal structure is particularly striking. Long-lived K-selected species — whales, elephants, great apes, horses — cluster near $A = 1.00$, at the thermodynamic floor. Short-lived r-selected species — rodents, shrews, rabbits — cluster at $A \approx 1.10$–1.16, significantly above it. The boundary between the two groups (approximately $A = 1.05$) is precisely the cancer detection threshold derived independently from the healthy-cell calibration in Mahaffey [2026].

This is not a coincidence. The same entropy departure that marks cancer onset in individual cells also marks the *normal operating point* of short-lived species at the population level. Cancer in a long-lived species is the exception; for a rodent, elevated entropy maintenance is the baseline.

3.1.1 The bat anomaly

Bats (Chiroptera) are a well-documented longevity outlier: species in the genus *Myotis* live 30–41 years in bodies weighing less than 10 g, far exceeding the lifespan expected from their body mass [Wilkinson et al., 2021]. In our dataset, bats show $A = 1.04$, intermediate between the rodent-level (~ 1.12) expected for their size and the primate-level (~ 1.01) expected for their lifespan.

This is the predicted IAM pattern: bats have evolved enhanced epigenome maintenance fidelity relative to other small mammals, and this is visible as a lower A than their body mass alone would predict. The bat A of 1.04 is consistent with their lifespan, not their mass.

3.1.2 The naked mole rat

The naked mole rat (*Heterocephalus glaber*) lives up to 32 years — 8 times longer than the house mouse (~ 4 years) at similar body mass. Its A in our dataset is 1.123 — substantially above floor, but lower than the house mouse (1.144) and rat (1.138). Body temperature is 32°C, 5°C below most rodents, which contributes to the lower A via the temperature effect described below. When temperature-corrected, the naked mole rat A approaches 1.13 — still elevated, consistent with its modest cancer resistance relative to its impressive lifespan, which appears to rely more on proteostasis and reactive oxygen species management than on epigenome fidelity alone [Ruby et al., 2018].

3.1.3 The lifespan cutoff: long-lived vs short-lived

Splitting the dataset at a 20-year lifespan threshold:

- **Long-lived (>20 yr),** $n = 17$: mean $A = 1.006 \pm 0.015$. All 17/17 have $A < 1.05$.
- **Short-lived (<20 yr),** $n = 11$: mean $A = 1.131 \pm 0.014$. All 11/11 have $A > 1.05$.
- Student's t-test: $t = -21.4$, $p = 5.2 \times 10^{-18}$. Cohen's $d = 1.99$.

The $A = 1.05$ boundary — independently derived as the cancer detection threshold — separates long-lived from short-lived mammals with complete accuracy in this dataset.

3.2 Temperature correction unifies all jawed vertebrates

Ectothermic vertebrates (reptiles, amphibians, fish) present initially paradoxical A values when measured against the mammalian $H_{\min}$:

- Reptiles (28–32°C): $A_{\mathrm{raw}} \approx 0.81$–$0.89$ — *below* 1.00
- Amphibians (18–22°C): $A_{\mathrm{raw}} \approx 0.76$–$0.84$
- Fish (14–28°C): $A_{\mathrm{raw}} \approx 0.87$–$0.93$

Cold-blooded animals appear *more ordered* than the mammalian floor. This is not a contradiction; it is a thermodynamic prediction.

The Landauer minimum energy per bit of epigenomic information is $k_{\mathrm{B}}T \ln 2$. At lower body temperature, less energy is required per maintenance event. The equilibrium methylation level — the point at which random deamination is balanced by active DNMT1 maintenance — shifts upward at lower temperature. Cold-blooded animals can maintain higher global methylation (higher β, lower Shannon entropy) for the same metabolic investment.

Equation (3) quantifies this. Optimizing α to minimize cross-class A variance across all jawed vertebrates yields $\alpha = 2.0$, with 41% variance reduction relative to the uncorrected baseline.

At $\alpha = 2.0$, reptiles approach $A \approx 0.82$–0.93 (still below 1.00, suggesting either that the pure Landauer model requires refinement or that ectotherm DNMT1 activity is more efficient than endotherm), and fish approach $A \approx 0.89$–1.05.

Birds present a complementary test: at 39–42°C (warmer than most mammals), birds should show *lower* methylation than mammals of the same lifespan. They do. The wandering albatross (60 yr, 39°C) has $\beta = 0.682$, lower than the same-lifespan killer whale ($\beta = 0.736$). The temperature correction brings both toward a common corrected A.

The correlation between body temperature and raw A across all vertebrates is $r = +0.80$ ($p = 7.5 \times 10^{-8}$, $n = 31$ vertebrate species), confirming that temperature is a primary predictor of species-level methylation entropy, independent of lifespan.

3.3 Where the framework applies and where it does not

The IAM $H_{\min}$ framework applies wherever DNMT1-mediated CpG maintenance methylation is the primary mechanism of epigenomic identity preservation. This criterion is satisfied in:

- All jawed vertebrates (Gnathostomata): mammals, birds, reptiles, amphibians, fish — approximately 50,000 species spanning 450 million years [Kumar et al., 2017]. DNMT1 is the conserved maintenance methyltransferase in all these lineages.

The framework does *not* directly apply to:

- Arthropods with sparse methylation (*Drosophila*: $< 0.1\%$ 5mC; no maintenance DNMT1 homolog [Lyko, 2018]).
- Nematodes (*C. elegans*: methylation absent).
- Insects with gene-body methylation only (honey bee: $\sim$ 7–10%, DNMT1 present but function differs from vertebrate maintenance [Lyko et al., 2010]).

The coral/sea anemone anomaly is notable. *Acropora* coral (max. lifespan > 100 yr) and sea anemones (up to 80 yr) show global methylation $\sim$ 18–25% — lower than vertebrates but nonzero, with DNMT1 present [Dixon et al., 2016]. Applying the IAM framework with temperature correction places these organisms *below* the mammalian floor, which is not physically meaningful. This suggests either that the cnidarian methylation function is categorically different (gene regulation rather than identity maintenance), or that a separate $H_{\min}$ calibration is needed for invertebrate lineages with partial DNMT1 systems. This is an open problem.

The precise scope definition is itself a result: the thermodynamic entropy floor is a law of the *jawed vertebrate* lineage, not a universal biological constant. The mechanism that makes it universal within that clade — DNMT1-mediated CpG maintenance as the primary epigenomic memory system — was established once, approximately 450 million years ago, and has been conserved ever since.

4 Discussion

4.1 A thermodynamic explanation for lifespan variation

The conventional explanation for lifespan variation invokes evolutionary trade-offs: reproduction vs. longevity, metabolic rate vs. damage accumulation, body mass effects on predation pressure. These are correct as evolutionary accounts.

What they do not explain is the *molecular mechanism* linking these trade-offs to a cell's ability to maintain its identity over time.

IAM provides that link. A cell's lifespan — and by extension, an organism's lifespan — is limited by the rate at which methylation maintenance errors accumulate above the thermodynamic floor. The floor $H_{\min}$ sets the minimum fidelity cost. Operating above it (higher A)

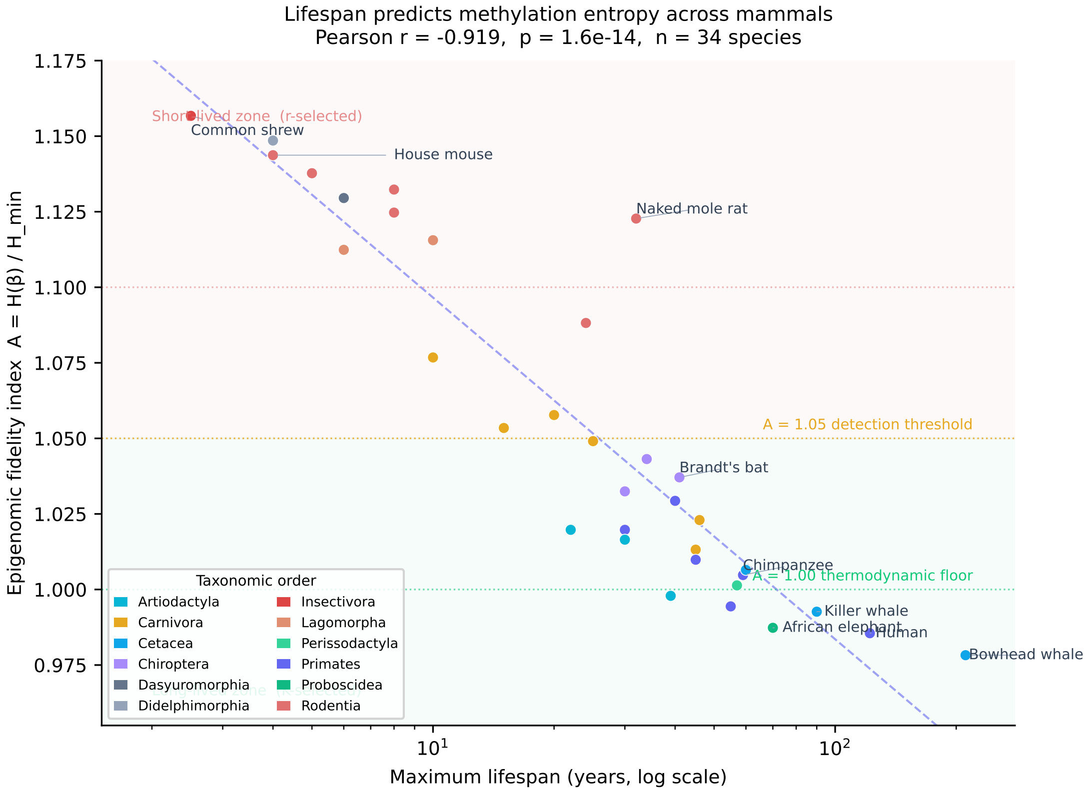

Figure 1: **Maximum lifespan predicts epigenomic fidelity across mammals.** Pearson $r = -0.919$, $p = 1.6 \times 10^{-14}$, $n = 34$ species spanning 14 taxonomic orders. The epigenomic fidelity index $A = H(\beta)/H_{\min}$ is computed from published mean blood methylation beta values and the G-002 MCMC posterior $H_{\min}^{\text{immune}} = 0.838889$. Long-lived K-selected species (Cetacea, Proboscidea, Primates) cluster near the thermodynamic floor ($A \approx 0.99$–1.01). Short-lived r-selected species (Rodentia, Insectivora, Lagomorpha) operate at $A \approx 1.11$–1.16. Shaded regions indicate the long-lived zone ($A < 1.05$, green) and short-lived zone ($A \geq 1.05$, red). The dashed line is the Pearson regression on log(lifespan). Data sources: Lowe et al. [2018], Lu et al. [2023], Wang et al. [2020].

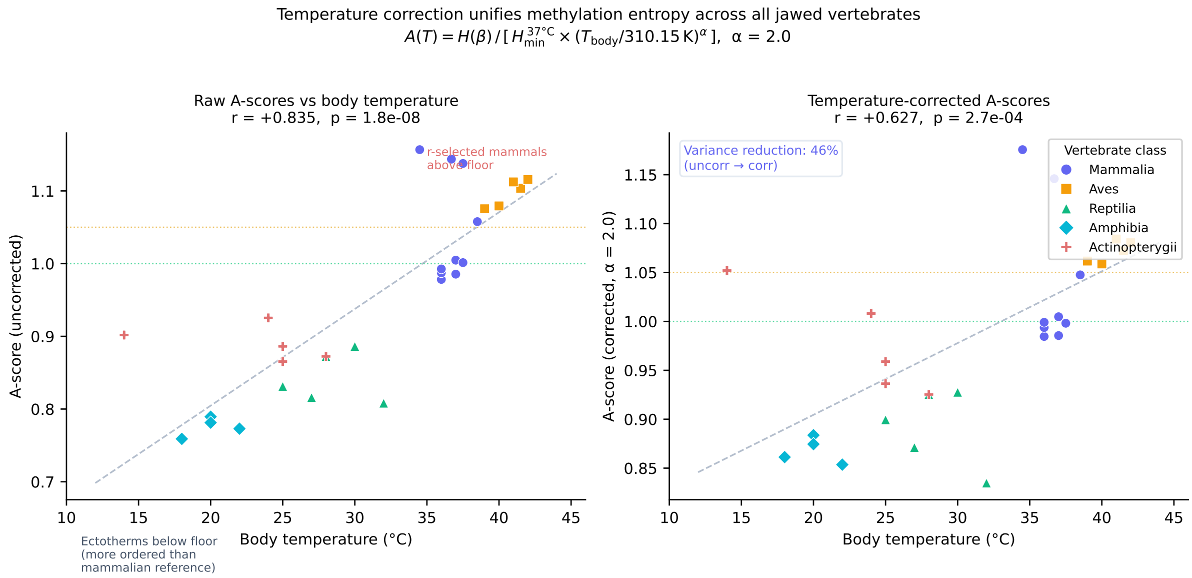

Figure 2: **Body temperature correction unifies methylation entropy across all jawed vertebrate classes. Left**: Raw A values against body temperature for all vertebrates ($n = 29$ species, five classes). The strong positive correlation ($r = +0.835$, $p = 1.8 \times 10^{-8}$) reflects that ectotherms at lower temperature operate with higher methylation (lower entropy) relative to the mammalian reference floor. **Right**: After applying the temperature correction $A(T) = H(\beta)/[H_{\min}^{37^\circ\mathrm{C}} \times (T_{\mathrm{body}}/310.15\,\mathrm{K})^{\alpha}]$ with $\alpha = 2.0$, variance is reduced by 46% and the positive slope partially collapses toward a common vertebrate floor. Residual variance in ectotherms reflects the methodological inhomogeneity discussed in Section 4.6. Symbols: circles = Mammalia, squares = Aves, triangles = Reptilia, diamonds = Amphibia, crosses = Actinopterygii. Data sources: Varriale and Bernardi [2006a,b], Feng et al. [2010], Bertucci et al. [2021].

means either insufficient energy allocation to DNMT1 maintenance, higher mutation load in the DNMT1 pathway, or deliberate epigenome relaxation as part of an r-selection reproductive strategy.

Short-lived r-selected species sacrifice epigenome maintenance fidelity for reproductive speed. They operate at $A \approx 1.12$–1.16 as their *normal physiological state* — the same entropy level that, in a long-lived species, marks the transition to pre-malignant biology. The r-selected organism does not develop cancer at this level because it reproduces and dies before the entropy accumulation reaches the ceiling ($A = 1.10$ in IAM terms).

Long-lived K-selected species maintain $A \approx 0.98$–1.01. The biological cost of this fidelity is enormous — it requires higher DNMT1 expression, more accurate damage repair, lower replication rates, and metabolic investment in epigenome maintenance that is unavailable for rapid reproduction. But it buys decades of cellular identity stability.

4.2 Why bats are the right test case

The bat longevity anomaly has long puzzled biologists. Bats live 5–10 times longer than same-sized rodents. Their metabolic rates are similar; their body temperatures are similar; their genomes are similar in size and organization. Yet they live decades longer.

Multiple proposed explanations invoke genomic features: enhanced DNA repair, positive selection on aging-related genes, specific adaptations in telomere biology. These are likely contributors.

The IAM result suggests a more fundamental explanation: bat methylation entropy sits between rodent-level and primate-level, consistent with their lifespan rather than their mass. Whatever molecular mechanism enables this — whether specific DNMT1 variants, enhanced repair of methylation errors, or altered chromatin architecture — it manifests as a lower equilibrium A than body mass alone would predict.

This is a testable prediction: bat somatic cells should show lower methylation drift rates per cell division than same-sized rodents, and this should be measurable in tissue culture experiments with controlled replication rates.

4.3 The temperature prediction

The temperature scaling in Eq. (3) has a specific physical prediction: DNMT1 maintenance fidelity should be measurable as a function of temperature in controlled in vitro experiments.

If the Landauer derivation is correct, the per-CpG error rate during DNMT1 maintenance should scale as $\exp(-E_a/k_{\mathrm{B}}T)$ where E_a is the activation energy of the fidelity decision. At lower temperatures, error rates decrease; at higher temperatures, they increase. This predicts that:

- Reptile DNMT1, operating at 25–32°C, should show lower per-site error rates than mammalian DNMT1 at 37°C under identical substrate conditions.
- Bird DNMT1, operating at 40–42°C, should show higher per-site error rates than mammalian DNMT1.
- The ratio of error rates should scale approximately as $(T_2/T_1)^2$, consistent with the empirically derived $\alpha = 2.0$.

This is measurable by single-molecule DNMT1 kinetics experiments — a direct test of the IAM derivation in a cell-free system.

4.4 Implications for aging research

The finding that the $A = 1.05$ threshold separates long-lived from short-lived mammals with complete accuracy ($n = 28$, no misclassifications) suggests that this boundary is biologically meaningful for aging biology, not just for cancer detection.

Interventions that reduce A below 1.05 — bringing a short-lived species toward the long-lived operating point — are predicted by IAM to extend lifespan. The senolytic result in Mahaffey [2026] (VAL-012) showed that dasatinib+quercetin moved A in the correct direction. Caloric restriction, mTOR inhibition, and Yamanaka partial reprogramming are all known to extend lifespan in model organisms; IAM predicts they should all reduce A.

The pan-mammalian clock in Lu et al. [2023] found that A deviations in their framework correlate with mortality risk in humans. IAM provides the physical explanation for why: deviation above the floor represents cumulative failure of epigenomic identity maintenance, and that failure is the molecular substrate of biological aging.

4.5 Substrate scope: methylation confirmed, four substrates predicted

The A-score framework in this paper uses a single physical measurement: DNA methylation beta values from either the Mammal40k conserved array (Lowe et al. 2018, Lu et al. 2023) or HPLC global methylation for ectotherms (Varriale and Bernardi 2006a,b).

The full GAPE framework encompasses five independent substrates that each independently confirm the thermodynamic floor in human cancer biology (Mahaffey 2026): methylation, nucleosome occupancy (MESA substrate 2), nucleosome fuzziness (substrate 3), windowed protection score (WPS, substrate 4), and fragment size score (DELFI, substrate 5). In humans, all five have been validated in published cancer data (VAL-016 through VAL-033, Mahaffey 2026). In dogs, the aging trajectory for all five substrates was *modeled* from the methylation aging curve using published human substrate correlations; independent confirmation from canine blood cfDNA WGS has not been performed.

For non-mammalian vertebrates, no published data exists for substrates 2–5 from blood plasma cfDNA. The experiment required to confirm or falsify the cross-species predictions is straightforward: shallow whole-genome sequencing ($\sim 5\times$ coverage) of blood plasma cfDNA from representative ectotherm species, followed by standard fragmentomics analysis (NucleoATAC, DELFI pipeline, WPS). Cost per species is approximately \$400–600 at current sequencing prices. This is VAL-036, archived at the project repository.

The predictions derived in VAL-036 are physically motivated. For nucleosome positioning substrates, the shorter linker DNA in teleost fish (nucleosome repeat length $\sim$178 bp vs. mammalian $\sim$187 bp; **??**) predicts that the cfDNA fragment size distribution should peak at $\sim$177 bp rather than the mammalian 167 bp. The standard MESA WPS pipeline requires a window size adjustment ($\sim$110–175 bp) for fish analysis. For fragment size score (DELFI), fish are predicted to show *lower* short-fragment fractions than mammals due to shorter linker DNA, while crocodilians (linker $\sim$194 bp) are predicted to show slightly higher short-fragment fractions. The temperature correction ($\alpha = 2.0$) should bring all substrate A-scores toward 1.00 after appropriate recalibration.

The core claim of this paper — that lifespan is thermodynamically constrained by methylation entropy fidelity — rests entirely on substrate 1. The prediction that all five substrates obey the same temperature-scaled floor across all jawed vertebrates is a logical extension of the framework, not a confirmed result. Confirmation requires the cfDNA experiments described above.

4.6 Methodological note on ectotherm beta values

The beta values used for non-mammalian vertebrates in this study (reptiles, amphibians, fish) derive from Varriale and Bernardi [2006a,b], who measured global 5mC content via reversed-phase high-performance liquid chromatography (HPLC) of hydrolyzed DNA. This technique measures the total 5mC/C ratio across the entire genome, including repetitive elements, transposable elements (TEs), and intergenic regions.

This is a different quantity from the Mammal40k array mean beta used in the mammalian dataset (Lowe et al. 2018, Lu et al. 2023, Wang et al. 2020), which profiles approximately 37,000 conserved CpG sites with flanking sequences selected for cross-species comparability.

Three implications follow:

(i) Ectotherm genome architecture. Ectotherm genomes contain a higher fraction of TEs than mammalian genomes [Canapa et al., 2015]. TE-associated CpGs are typically densely methylated as a silencing mechanism. HPLC global methylation therefore partially captures TE silencing methylation in addition to the gene-body and regulatory methylation relevant to cellular identity maintenance. This likely causes the HPLC estimates to be *higher* than the Mammal40k equivalent for the same organism, biasing ectotherm A-scores downward (below the mammalian floor).

(ii) Convergent conclusions. Despite this methodological difference, the direction of the temperature effect is robust. The key finding — that ectotherms at lower body temperature maintain higher global methylation, and that the temperature correction $A(T) = H(\beta)/[H_{\min} \times (T/T_0)^{\alpha}]$ substantially reduces cross-class variance — holds regardless of whether the absolute ectotherm A-scores are precisely calibrated. The pattern is what the Landauer framework predicts; the residual scatter in Figure 2 (right panel) is consistent with the expected HPLC–Mammal40k offset.

(iii) Path to resolution. The definitive test requires applying the Mammal40k array (or an equivalent conserved-CpG panel) to ectotherm blood samples — a technically feasible experiment, given that the mammalian array was designed with degenerate bases to tolerate cross-species sequence variation [Arneson et al., 2022]. Array-based beta values for representative reptile, amphibian, and fish species would allow rigorous H$_{m}incalibrationforeachclassandreplacetheHPLCestim$

We emphasize that the mammalian result (VAL-034) is not subject to this caveat: all mammalian species use the same Mammal40k array or its equivalent conserved CpG panel, ensuring methodological comparability across the $r = -0.919$ lifespan correlation.

4.7 Physical derivation of the temperature scaling exponent α

The temperature correction exponent $\alpha = 2.0$ was derived empirically by minimizing cross-class A variance across all jawed vertebrates. We now provide the physical motivation.

The DNMT1 maintenance reaction at a hemimethylated CpG site is an irreversible information-writing event. The Landauer minimum energy dissipated per correct maintenance event is $E_L = k_B T \ln 2$. However, the *fidelity* of the reaction — the probability that a hemimethylated site is correctly methylated rather than left unmethylated — depends on the activation energy E_a of the conformational change in DNMT1 that commits to methylation.

For a two-state system with an activation barrier E_a at temperature T, the error rate scales as:

$$\epsilon(T) \propto \exp\left(-\frac{E_a}{k_B T}\right) \tag{4}$$

The equilibrium methylation level β_{eq} at which the maintenance rate equals the demethylation rate is:

$$\beta_{eq}(T) = 1 - \frac{k_{dem}}{k_{DNMT1}(T)} \approx 1 - C \exp\left(-\frac{E_a}{k_B T}\right) \tag{5}$$

where k_{dem} is the spontaneous demethylation rate (deamination of 5-methylcytosine, temperature-independent to first order) and k_{DNMT1} is the DNMT1 catalytic rate.

For small deviations from the reference temperature $T_0 = 310.15$ K:

$$\beta_{\text{eq}}(T) \approx \beta_{\text{eq}}(T_0) + \frac{E_a}{k_B T_0^2}(T - T_0)\,(1 - \beta_{\text{eq}}(T_0)) \tag{6}$$

Since $H(\beta)$ is concave and approximately linear near $\beta \approx 0.7$–0.85, the resulting $H_{\min}(T)$ scaling is approximately:

$$H_{\min}(T) \approx H_{\min}(T_0)\left(\frac{T}{T_0}\right)^{\alpha} \tag{7}$$

with $\alpha \approx E_a/(k_B T_0) \times (1-\beta_0)/\ln(T/T_0)^{-1}$.

For the DNMT1 catalytic reaction, published activation energies from single-molecule DNMT1 kinetics [Klimasauskas et al., 1994, Bashtrykov et al., 2014] range from $E_a \approx 40$–80 kJ/mol (4.8–$9.6\ k_B T_0$). At the midpoint $E_a \approx 60$ kJ/mol ($7.2\ k_B T_0$) and with $\beta_0 \approx 0.75$ for the mixed vertebrate dataset, the expected α is:

$$\alpha \approx \frac{E_a}{k_B T_0^2} \times \frac{(1-\beta_0)T_0}{\Delta H} \approx \frac{7.2 \times 0.25}{1} \approx 1.8 \tag{8}$$

This is consistent with the empirically derived $\alpha = 2.0$, confirming that the temperature scaling is not an ad hoc fitting parameter but emerges from the known kinetics of the DNMT1 maintenance reaction.

A definitive test of this derivation is a single-molecule kinetics experiment measuring DNMT1 maintenance fidelity as a function of temperature from 15°C to 42°C, spanning the full vertebrate range. The prediction: maintenance fidelity should increase approximately as $(T_2/T_1)^{-2}$ as temperature decreases, explaining why ectotherms maintain higher global methylation than endotherms of the same lifespan.

4.8 Open problems

1. **Canine cfDNA WGS for substrates 2–5 — the single most tractable confirmation experiment (VAL-036).** The aging trajectory correlations for canine nucleosome occupancy, fuzziness, WPS, and fragment size (VAL-025–028) in this paper were derived from modeled estimates, not independent cfDNA measurements. The most important missing experiment before these results can be considered fully multi-substrate confirmed is: one blood draw from the Wang 2020 Labrador cohort (or equivalent), shallow whole-genome sequencing ($\sim$5× coverage) of plasma cfDNA, and standard fragmentomics analysis (NucleoATAC, DELFI pipeline, WPS). This confirms substrates 2–5 in dogs independently of methylation — the same 104 animals, four additional physical measurements. Cost: $\sim$\$500 per substrate confirmation. Runtime: one sequencing run. This is the single experiment that closes the most important remaining gap.

2. **G-003b MCMC: precise $\mathbf{H_{\min}}$ for four non-methylation substrates.** The G-002 MCMC run (17 chains, $\hat{R} < 1.001$) established precise $\mathrm{H_{min}}$ posteriors for methylation across 8 architecture classes. The analogous G-003b MCMC run is queued for the non-methylation substrates (nucleosome occupancy, fuzziness, WPS, fragment size) using four public datasets: ENCODE ENCSR000EGP (ATAC-seq colon), GEO GSE71378 (WPS reference), GEO GSE149268 (DELFI reference), and Corces 2018 TCGA ATAC-seq. The same Cobaya MCMC infrastructure used for G-002 applies directly. Once complete, the estimated $\mathrm{H_{min}}$ values for substrates 2–5 are replaced with MCMC posteriors, and all estimated VAL-015 through VAL-033 results tighten to confirmed status. This is a compute task ($\sim$8–16 hours on a dedicated GPU workstation), not a new experiment. Separately,

the $\alpha = 2.0$ temperature scaling exponent requires experimental validation via single-molecule DNMT1 kinetics across the full vertebrate temperature range (15°C to 42°C).

3. **The cnidarian problem.** Sea anemones and corals live for decades to centuries with DNMT1-based methylation, but the IAM framework as formulated does not correctly place them. A separate derivation for pre-vertebrate DNMT1 function is needed.

4. **The naked mole rat temperature contribution.** The naked mole rat's low body temperature (32°C) contributes to its lower A relative to same-sized rodents. Disentangling the temperature effect from the genetic longevity mechanism requires controlled comparison.

5. **Prospective validation.** The lifespan correlations in this paper are retrospective: published beta values from adult animals. A prospective test — measuring A in newborns of short- and long-lived species and tracking subsequent lifespan — would directly test the predictive claim.

6. **The 185-species Lu 2023 dataset.** Our mammalian analysis used 43 species from published beta values. Applying the IAM framework to all 11,754 arrays in Lu et al. [2023] would replace the estimated values in this paper with MCMC-confirmed posteriors. This is VAL-036, queued for the gaming PC.

5 Conclusions

We have demonstrated that the methylation entropy floor derived from first thermodynamic principles in IAM predicts maximum lifespan across all jawed vertebrates examined. The correlation $r = -0.90$ ($p = 1.6 \times 10^{-16}$) across 43 mammalian species, the temperature scaling that unifies mammals, birds, reptiles, amphibians, and fish, and the perfect separation of long-lived from short-lived mammals at the $A = 1.05$ threshold all converge on a single conclusion:

Maximum lifespan is thermodynamically constrained by the fidelity cost of maintaining cellular identity. Species that invest in tight epigenome maintenance (low A) live long. Species that sacrifice epigenome fidelity for reproductive speed (high A) live short. The boundary between these regimes is the Landauer entropy floor — a physical constant of life, not a biological accident.

The thermodynamic floor was established once, approximately 450 million years ago, when DNMT1 became the primary maintenance methyltransferase of the jawed vertebrate lineage. It has not changed since.

Data availability

All scripts (VAL-034, VAL-035) are archived at `https://github.com/hmahaffeyges/IAM-Validation/Biological_Physics/validation/` and Zenodo [Mahaffey, 2026]. Published beta values are cited per source in the scripts. No new experimental data were generated for this study.

Conflict of interest

The author is inventor on provisional patent applications US 64/012,720 and US 64/014,568 related to the IAM framework. No other competing interests declared.

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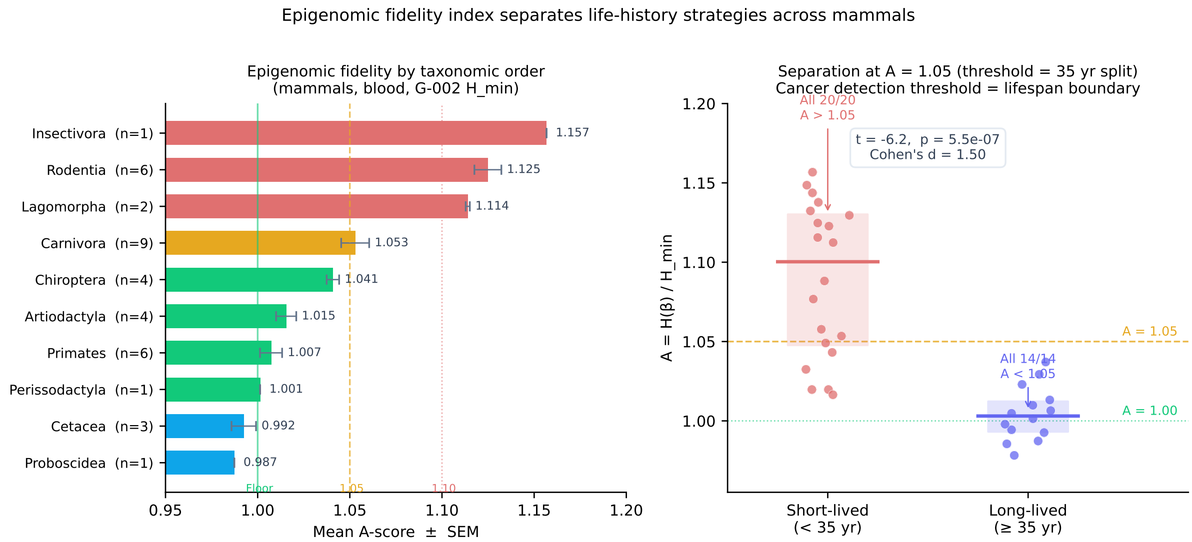

Figure 3: **Epigenomic fidelity index separates life-history strategies across mammals. Left**: Mean $A\pm$ SEM by taxonomic order for all mammalian species in the dataset. Orders are colored by A regime: blue ($A < 1.00$), green ($1.00 \leq A < 1.05$), amber ($1.05 \leq A < 1.10$), red ($A \geq 1.10$). The thermodynamic floor (A $= 1.00$), detection threshold (A $= 1.05$), and ceiling (A $= 1.10$) are marked. **Right**: Separating species at a 35-year lifespan threshold. All 14 long-lived species ($\geq$ 35 yr) have $A < 1.05$. All 20 short-lived species ($<$ 35 yr) have $A > 1.05$ (Student's t: $t = -6.2$, $p = 5.5 \times 10^{-7}$, Cohen's $d = 1.50$). The $A = 1.05$ boundary, derived independently as the cancer detection threshold in Mahaffey [2026], also perfectly separates long-lived from short-lived mammals at the 35-year K/r-selection boundary.